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Entd in L.B. Suppl. 1. p. 132.

An Argument
for divine providence
taken from the constant regularity
observed in the Births of Both Sexes.

By Dr Arbuthnot.

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Among innumerable footsteps of Divine providence to be found in the works of Nature, there is a very remarkable one, to be observed, in the exact Ballance that is maintained, between the numbers of men and women; for by this means it is provided, that the species may never fail, nor perish, since every male may have its female, and of a proportionable age. This æquality of males and females is not the effect of chance but divine providence, working for a good end such I thus demonstrate,

Let there be a die of two sides, M. and F, (wch denote cross and pile), now to find all the chances of any determined number of such dice, let the Binome $M+F$ be raised to ye power whose exponent is the number of dice given; the coefficients of the terms will shew all the chances sought. For Example, in two dice of two sides $M+F$ the Chances are $M^2+2MF+F^2$ that is one chance for M double, one for F double, and 2 for M single and F single; in four such dice, there are chances $M^4+4M^3F+6M^2F^2+4MF^3+F^4$, that is one chance for M quadruple, one for F quadruple, four for triple M, and single F, four for single M and triple F, and 6 for M double, and F double; and universally, if the number of dice be n, all their chances will be expressed in this series

$$M^n + \frac{n}{1} \times M^{n-1}F + \frac{n}{1} \times \frac{n-1}{2} \times M^{n-2}F^2 + \frac{n}{1} \times \frac{n-1}{2} \times \frac{n-2}{3} \times M^{n-3}F$$

It appears plainly, that when the number of dice is even there are as many M's as F's in the midle term of this series and in all the other terms, there are most M's or most F's.

If therefore a man undertake with an even number of dice to throw as many M's, as F's he has all the terms but the middle term against him; and his ~~[illegible]~~ Lot, is to the sum of all the Chances, as the coefficient of the middle term is to the power of 2 raised to an exponent equall to the number of dice so in two dice his Lot is $\frac{2}{4}$, or $\frac{1}{2}$, in three dice $\frac{6}{16}$ or $\frac{3}{8}$, in six dice $\frac{20}{64}$, or $\frac{5}{16}$, in eight $\frac{70}{256}$, or $\frac{35}{128}$ &c.

To find this middle term in any given power or number of dice, Continue the series $\frac{n}{1} \times \frac{n-1}{2} \times \frac{n-2}{3}$ &c: till the number of terms are equall to $\frac{1}{2}n$. for Example the coefficient of the middle term of the 10th power is $\frac{10}{1} \times \frac{9}{2} \times \frac{8}{3} \times \frac{7}{4} \times \frac{6}{5} = 252$, the 10th power of 2 is 1024, if therefore A undertakes to throw with 10 dice in one throw, an equal number of M's and F's he has 252 chances out of 1024, for him, that is his lot is $\frac{252}{1024}$ or $\frac{63}{256}$, which is less then $\frac{1}{4}$

It will be easy by the help of Logarithms, to extend this Calculation to a very great number, but that is not my present design; it is visible

from what has been said, that with a very great num-
-ber of dice, A's Lot would became very small; and
consequently (supposing M to denote Male, and F female) the
in the vast number of Mortals, there would be but a small
part of all the possible chances, for its happening at
&c:) any assignable time, that an equall number
of Males and females should be born.

It is indeed to be confessed that this æquality
of males and females is not Mathematical but
Physicall, wch alters much the foregoing cal-
-culation, for in this case. the middle term will
not exactly give A's chances, but his chances will
take in some of the terms next the middle one, and
will lean to one side or the other. But it is very
improbable, (if mear chance-governed,) that they would never
reach as far as the extremities. But this event
is wisely prevented by the wise æconomy of Nature
and to judge of the wise contrivance, we must ob-
-serve that the externall accidents, to wch Males are
subject (who must seek their food with danger) do
make a great havock of them, and that this
loss exceeds far that of the other sex, occasioned
by diseases incident ~~proper~~ to it, as experience convinces
us. to repair that loss provident Nature by the disposal
of its wise Creator, brings forth more Males than
females. and that in almost a constant proportion
this appears from the annexed Tables, wch contain observations
of 82 years of the Births in London. now to reduce
the whole to a calculation I propose this,

Problem. A lays against B, that every year there shall be born more males than females. To find A's lot, or the value of his expectation

It is evident from what has been said that A's lot for each year is less than $\frac{1}{2}$ (but that the Argument may be stronger) let his lot be equal to $\frac{1}{2}$ for one year. if he undertakes to do the same thing 82 times running his lot will be $\frac{1}{2^{82}}$ wch will be found easily by the Table of Logarithms, to be $\frac{1}{4836\,0000\,0000\,0000\,0000\,00000}$. But if A wager with B. not only that the Number of Males shall exceed that of females, every year, but that this excess shall happen in a constant proportion, and the difference lye within fix'd limits; and this not only for 82 years but for ages of ages, & not only at London, but all over the world (wch is highly probable fact, designed that every male may have a female of the same country and suitable age.

then A's chance will be near an infinitely small quantity at least less than any assignable fraction. from whence it follows, that it is Art not chance that governs.

There seems no more probable cause to be assigned in Physicks for this equality of the Births, except that in our first parents seed, there were at first formed an equall number of both sexes.

Scholium. From hence it follows that Polygamy is contrary to the law of Nature and justice, and to the propagation of human race. for where males and females are in equall number. if one man takes 20 wives 19 men must live in Celibate; wch is repugnant to the design of nature. nor is it probable that 20 women will be so well impregnated by one man as by 20.

	Christened			Christened	
Anno	Males	Females		Males	Females
1629	5218	4683	56	3668	3382
30	4858	4457	57	3396	3289
31	4422	4102	58	3157	3013
32	4994	4590	59	3209	2781
33	5158	4839	60	3724	3247
34	5035	4820	61	4748	4107
35	5106	4928	62	5216	4803
36	4917	4605	63	5411	4881
37	4703	4457	64	6041	5681
38	5359	4952	65	5114	4858
39	5366	4784	66	4678	4319
40	5518	5332	67	5616	5322
41	5470	5200	68	6073	5560
42	5460	4910	69	6506	5829
43	4793	4617	70	6278	5719
44	4107	3997	71	6449	6061
45	4047	3919	72	6443	6120
46	3768	3395	73	6073	5822
47	3796	3536	74	6113	5738
48	3363	3181	75	6058	5717
49	3079	2746	76	6552	5847
50	2890	2722	77	6423	6203
51	3231	2840	78	6568	6033
52	3220	2908	79	6247	6041
53	3196	2959	80	6548	6299
54	3441	3179	81	6822	6533
55	3655	3349	82	6909	6744

	Christened			Christened	
Anno:	Males	Females	Anno:	Males	Females,
83	7577	7158	1700	7578	7061
84	7575	7127	1701	8102	7514
85.	7484	7246	1702	8031	7656
86	7575	7119	1703	7765	7683
87	7737	7214	1704	6113	5738
88.	7487	7101	1705	8366	7779
89	7604	7167	1706	7952	7417
90	7909	7302	1707	8379	7687
91	7662	7392	1708	8239	7623
92	7602	7316	1709	7840	7380
93	7676	7483	1710	7640	7288
94	6985	6647			
95	7263	6713			
96	7632	7229			
97	8062	7767			
98.	8426	7626			
99	7911	7452			

Numb. 1.

Dr Arbuthnots Argt.
to Prove the Equality
of Males & Females &c

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